

VINGA, Sweden

WMO: 025160

Lat: 57.6322N

Lon: 11.6048E

Elev: 59

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.3	20.5	9.4	8.9	21.7	13.4	10.9	24.1	42.2	37.5	38.7	38.8	14.1	50	0.764

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	73.8	66.0	71.6	65.3	69.6	63.7	68.0	72.2	66.2	70.6	64.5	68.9	9.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	97.2	70.9	64.3	90.4	69.4	62.5	84.8	67.5	32.4	72.3	31.0	70.9	29.7	69.0	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.3	34.0	30.6	17.4	77.1	5.4	2.5	13.5	79.0	10.3	80.4	7.3	81.8	3.3	83.7		
(5)			17.3	68.5	5.2	2.6	13.5	70.3	10.5	71.8	7.6	73.3	3.8	75.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	34.0	33.6	36.8	44.4	52.7	59.0	63.9	63.7	58.0	50.3	42.6	37.4
	DBStd	12.04	6.47	6.34	5.04	4.85	5.42	4.02	4.08	3.39	3.66	4.40	5.74	6.57
	HDD50	2239	496	459	408	182	27	0	0	0	1	51	225	390
	HDD65	6232	961	879	873	619	382	186	72	67	210	457	671	855
	CDD50	1554	0	0	0	13	112	270	430	424	241	60	4	0
	CDD65	73	0	0	0	0	2	7	37	26	1	0	0	0
	CDH74	57	0	0	0	0	4	3	40	10	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	15.6	17.6	16.4	14.7	13.1	13.1	13.7	12.9	14.3	16.7	18.0	18.6	18.2
Precipitation	PrecAvg	22.1	1.7	1.1	1.3	1.3	1.6	1.8	2.0	2.4	2.6	2.4	2.2	1.9
	PrecMax	28.5	3.7	3.1	2.8	2.7	4.1	3.8	5.0	6.1	6.9	6.1	4.2	5.1
	PrecMin	17.0	0.0	0.0	0.0	0.0	0.2	0.1	0.3	0.2	0.3	0.8	0.5	0.4
	PrecStd	3.0	1.0	0.7	0.8	0.7	0.8	0.9	1.2	1.4	1.3	1.2	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	44.4	50.8	64.3	72.5	73.7	77.8	75.5	68.9	59.7	54.8	48.6
		MCWB	44.4	41.8	44.9	53.0	61.2	62.8	66.9	67.6	62.3	56.5	53.4	46.5
	2%	DB	45.3	43.0	46.1	57.2	67.1	69.8	74.6	73.1	66.1	58.1	52.7	47.4
		MCWB	42.9	41.2	42.6	49.6	58.3	62.3	66.2	66.7	60.4	55.4	50.9	45.5
	5%	DB	43.6	41.8	43.9	53.5	63.3	67.3	72.0	71.3	64.4	57.0	51.5	46.3
		MCWB	41.6	40.2	41.4	47.9	56.5	60.9	65.5	65.9	60.1	54.2	49.5	44.6
	10%	DB	42.1	40.6	42.3	50.8	60.3	64.9	70.1	69.1	63.0	56.0	49.8	45.0
		MCWB	40.3	39.1	40.2	46.2	54.7	59.6	64.2	64.1	59.0	53.2	47.8	43.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	42.8	45.8	53.0	63.5	65.5	69.3	70.6	63.9	57.8	53.6	47.6
		MCDB	46.8	43.5	49.8	57.8	70.1	70.1	74.2	73.6	66.1	58.8	54.5	48.3
	2%	WB	43.6	41.5	43.6	50.3	60.2	63.5	67.8	68.7	61.8	56.1	51.3	46.3
		MCDB	44.8	42.4	45.9	54.5	65.6	67.8	72.6	72.0	63.9	57.6	52.4	47.0
	5%	WB	42.0	40.4	42.1	48.7	58.0	62.0	66.5	66.6	60.6	54.9	49.9	44.9
		MCDB	43.3	41.4	43.6	52.8	62.2	66.0	71.2	70.3	63.0	56.5	51.3	46.0
	10%	WB	40.7	39.5	40.9	46.9	55.6	60.6	64.9	64.7	59.3	53.9	48.3	43.7
		MCDB	42.0	40.5	42.2	50.3	59.4	64.6	69.6	68.8	62.2	55.7	49.8	44.9
Mean Daily Temperature Range	5% DB	MDBR	5.0	4.6	5.4	6.4	7.0	6.3	6.2	5.8	5.4	4.9	4.4	4.9
		MCDBR	5.4	4.2	6.9	10.6	11.8	9.5	8.9	8.2	6.7	4.5	4.2	4.4
		MCWBR	6.0	4.0	4.7	6.5	7.2	6.6	6.2	5.9	5.2	5.0	5.0	5.1
	5% WB	MCDBR	5.2	4.0	6.0	9.0	10.6	8.3	8.1	7.0	5.6	4.3	4.3	4.3
		MCWBR	5.7	4.0	4.8	5.8	7.2	6.4	6.3	6.0	5.3	5.3	5.1	4.9
Clear-Sky Solar Irradiance	taub	0.304	0.331	0.348	0.386	0.380	0.369	0.391	0.391	0.360	0.347	0.321	0.289	
	taud	2.254	2.314	2.353	2.299	2.396	2.454	2.410	2.406	2.477	2.413	2.302	2.208	
	Ebn at Noon	174	224	254	261	272	276	266	258	249	216	165	141	
	Edh at Noon	16	23	29	35	34	33	33	31	25	21	15	12	
All-Sky Solar Radiation	RadAvg	115	294	757	1279	1708	1896	1739	1357	918	422	153	79	
	RadStd	23	52	72	151	139	132	166	91	90	58	26	9	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	-223	-319	N/A	N/A

Nomenclature: See separate page